BEYOND THE THREADED PROGRAMMING MODEL ON REAL-TIME OPERATING SYSTEMS

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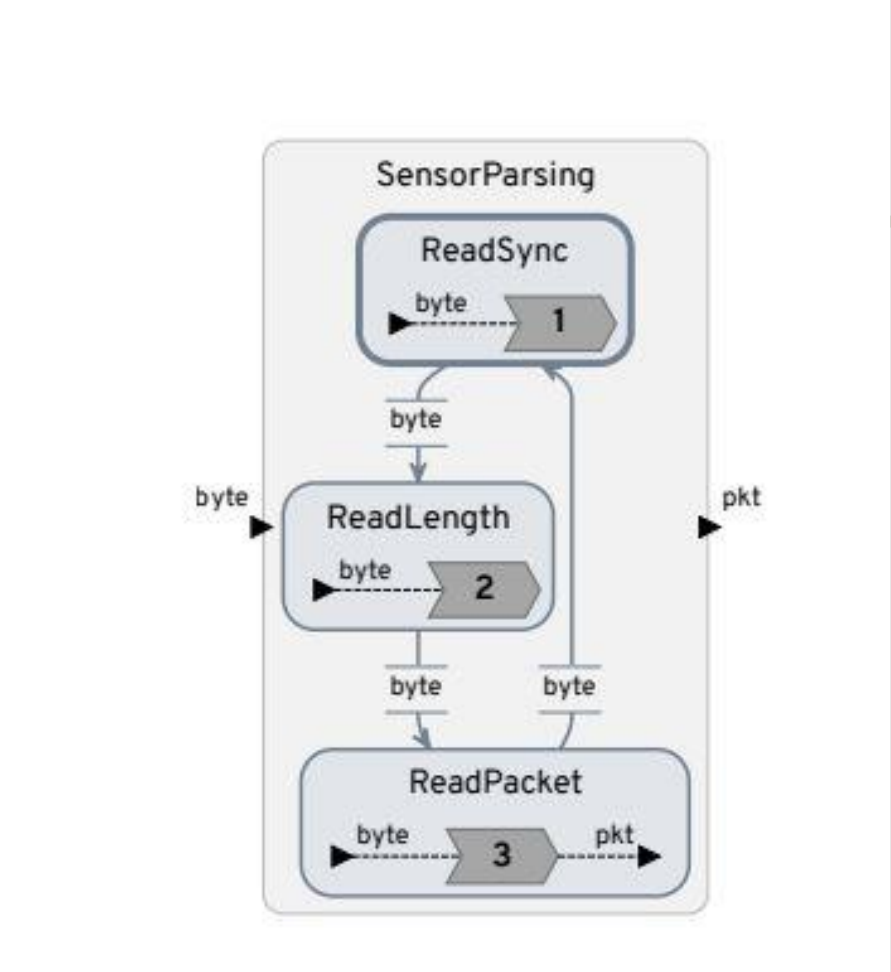
ABSTRACT

- **Embedded system**
- **RTOS : Threads support**
    - Zephyr
    - Free RTOS
- **CSP and Actors model**
- **Lingua franca**

LINGUA FRANCA

- Coordination language
- Reactor-Oriented Programming

REACTORS


SensorParsing
ReadSync
byte
1
byte
byte
ReadLength
pkt
byte
2
byte
byte
ReadPacket
byte
pkt
3

REACTORS

- target C
- main reactor {
- reaction(startup) {=
- printf("Hello World.\n");
- =}
- }

EMBEDDED DESIGN PATTERN

- 1. Button debouncing

```
target C;

reactor ButtonDebounce {
    preamble {=
        void * button_press_action; // Pointer to physical action
        void gpio_pin_isr() {
            lf_schedule(button_press_action, 0);
        }
    =}

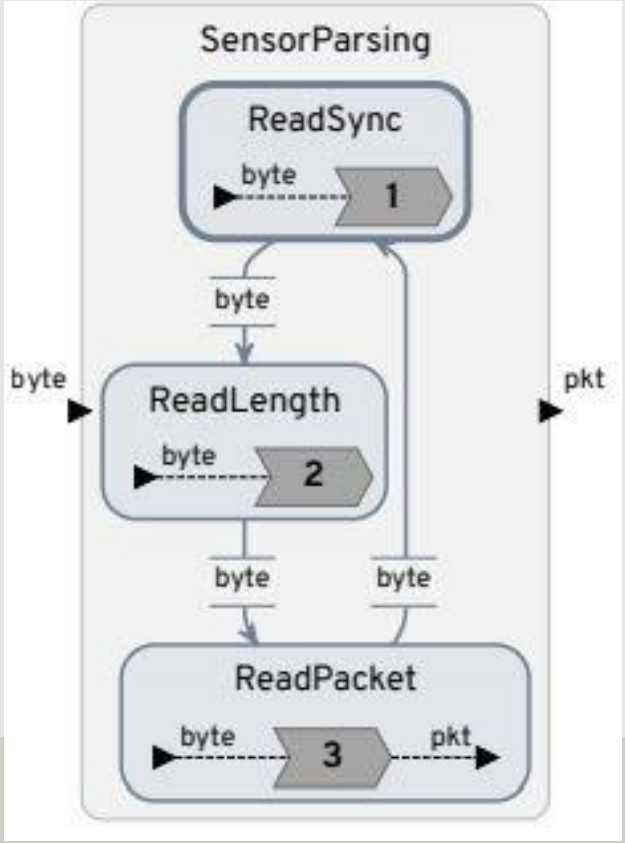
    physical action button_press(0, 10 msec)

    reaction(startup) -> button_press {=
        // Setup GPIO pin and register ISR
        ...
        // Make physical action available externally
        button_press_action = button_press;
    =}

    reaction (button_press) {=
        // Handle
    =}
}
```

EMBEDDED DESIGN PATTERN

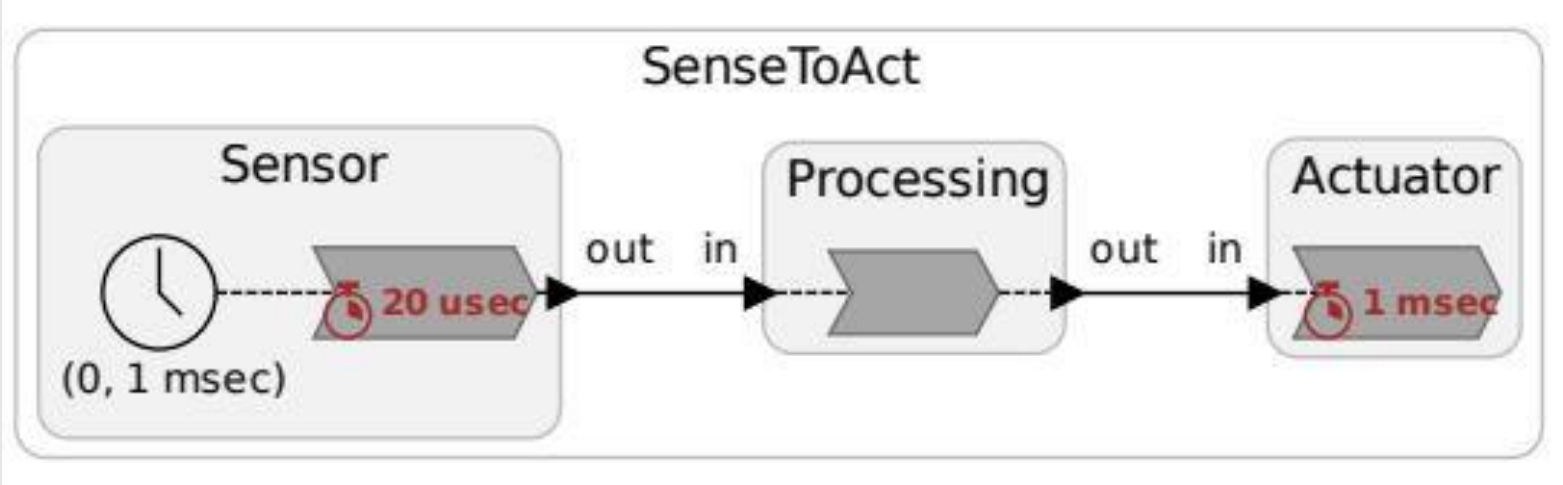
- 2. Sensor parsing




```
1  reactor SensorParser {
2    input byte:char
3    output pkt:sensor_pkt_t;
4
5    initial mode ReadSync {
6      reaction(byte) -> ReadLength {=
7        // Search for sync-byte ...
8        lf_set_mode(ReadLength);
9      =}
10   }
11
12   mode ReadLength {
13     reaction(byte) -> ReadPacket {=
14       // Read length field ...
15       lf_set_mode(ReadPacket);
16     =}
17   }
18
19   mode ReadPacket {
20     reaction(byte) -> pkt {=
21       // Read packet into p ...
22       lf_set(pkt, p);
23     =}
24   }
25 }
```

EMBEDDED DESIGN PATTERN

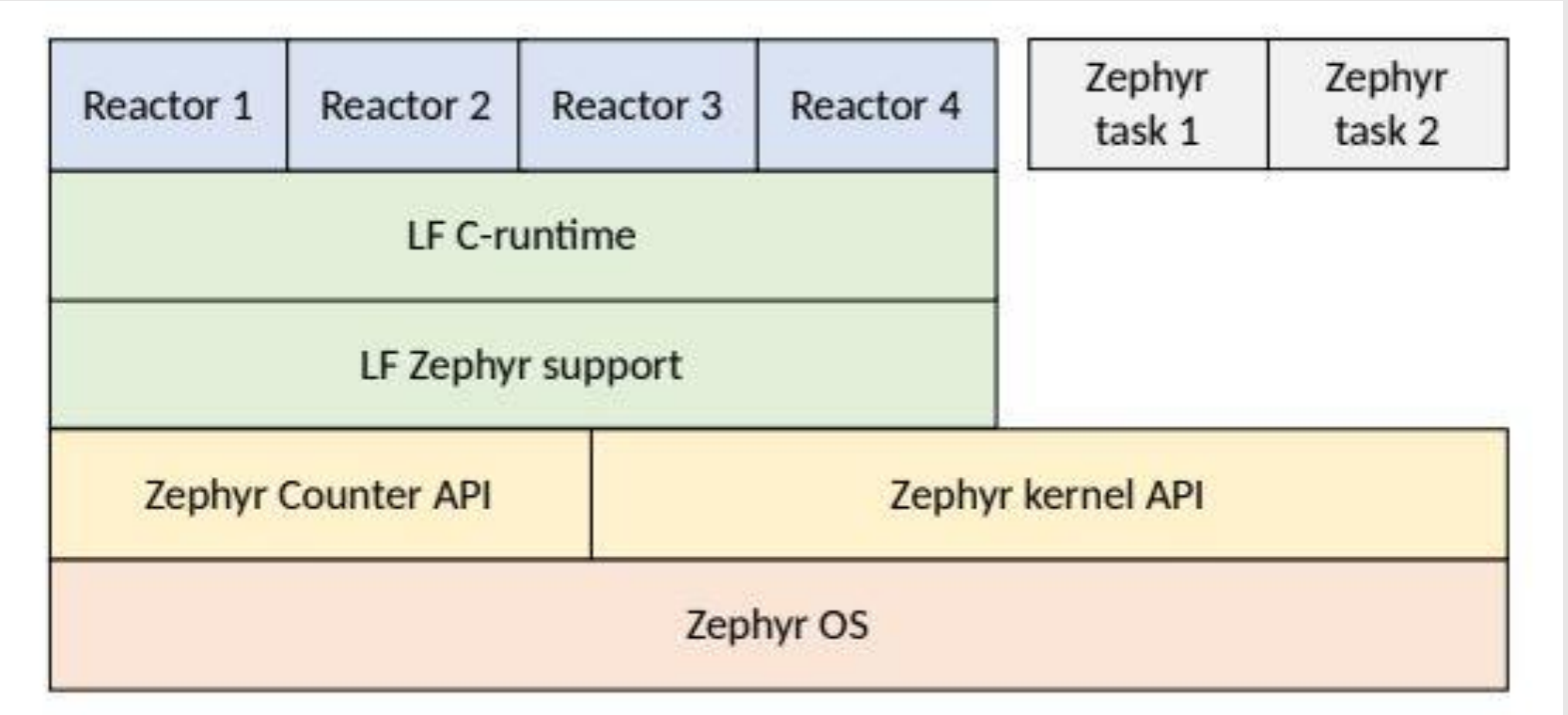
- 3. Control loops



DESIGN AND EVALUATION

- Zephyr OS

DESIGN AND EVALUATION

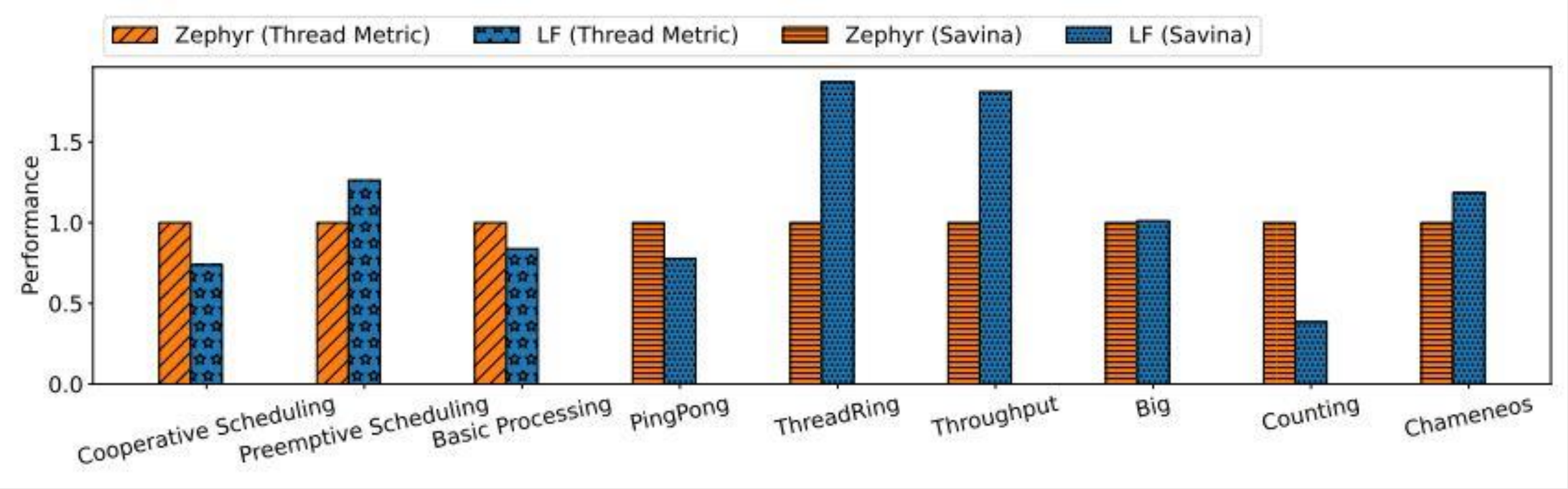


EVALUATION

- Memory footprint

	Runtime	Reactor	Reaction	Timer	Action	Mode	Port (In/Out)	Event
Code size	9.8KB(1KB)	96B	62B	88B	174B	36B	32B/16B	–
Data size	266B(170B)	72B	80B	60B	72B	28B	104B/40B	32B

PERFORMANCE



THANK YOU